Bit Operation Cost of Hold-out Key-Recovery Attacks Against Classic McEliece

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Abstract. We tally bit operations for a proposed hold-out key-recovery attack on Classic McEliece. In our *singleton hold-out* model, each relation kernel omits exactly one selected public column. We prove three relation-generation reductions: columns may use different vanishing orders, redundant binary derivative rows may be discarded, and injective zero-padding makes a rectangular relation matrix square without changing its right kernel. Modeling each column outside the identity block as half nonzero, an exhaustive singleton scan gives a width-64 relation-generation floor of $2^{112.35}$ bit operations. Under downstream assumptions, charging support-value enumeration and dense curve interpolation gives a conditional subtotal through interpolation of $2^{132.22}$ for mceliece348864. Neither figure is a complete key-recovery cost. Assuming higher-order derivative recovery adds one dimension at each order, Apon's bound shows that c initially guessed support values cannot isolate the correct binary-Goppa curve. Following the full-code alternative identified, we model finding at least $2t + 3 - c$ further correct values by matching unused public columns rather than by additional guessing. End-to-end attack complexity is not yet established: At target scale, we do not know how many relations higher-order derivative recovery requires, the success probability of binary kernel solving and public-column matching, or how to normalize the matched values using only public data.

Keywords: Classic McEliece · Hold-out Key Recovery · Bit Operations · Sparse Linear Algebra

1 Introduction

Classic McEliece derives its public code from a binary Goppa code and relies on the difficulty of recovering an equivalent decoding key [4,13]. Recent higher-order-vanishing (HOV) and hold-out attacks construct public spaces of multilinear relations that distinguish some Goppa-derived codes from random codes [8,11]. Vedenev subsequently described how suitable hold-out relations could enable higher-order derivative recovery on a hidden coefficient curve and, conjecturally for binary Goppa codes, recover a key [15]. The resulting route is structurally different from information-set decoding (ISD): it first computes a large public relation space and then enumerates and interpolates a small number of hidden curve anchors.

This paper consolidates that route into one conditional algorithm and one consistent arithmetic model. Some quantities follow exactly from the parameter search; others assume one sparse-solver attempt per relation batch or use explicit heuristics. The primary tables stop at the candidate-interpolation solve. The end-to-end reading assumes generic-rank rejection of incorrect guesses, matching unused public columns to extend the necessarily non-unique correct leaf, and an experimentally validated labelled-support endpoint. Optional phase synchronization and a conservative continuation envelope remain in the appendix.

Our contributions:

1. We give the binary hold-out operator, prove its exact point-block rank, strengthen Hermite soundness by charging systematic pivots at their full order $d-1$, and prove the heterogeneous-multiplicity and Lucas-compression reductions used to select cells.
2. We replace the rectangular operator by an injectively padded square operator with the same right kernel whenever $R_M \leq N$. We isolate the remaining small-field Krylov reliability assumption and give fixed-width and multiword-block prices.
3. We give executable-style pseudocode for relation generation, higher-order derivative recovery, candidate interpolation, and verified reconstruction, separating exact steps from solver, richness, and wrong-leaf heuristics.
4. We jointly optimize conditional subtotals for all five Classic McEliece parameter sets and report solver state, stored relations, and active retained state rather than conflating them.

Artifact. The public artifact repository contains a self-contained costing program `mccost.py`, and concise reproduction instructions for estimates in this paper. See: `https://github.com/mjosaarinen/mccost`

Terminology. An *anchor* is a held-out public column at which the attack performs *higher-order derivative recovery*; we use that name for the process throughout. Its output is a nested derivative filtration, called a local flag in the source literature. A *cell* fixes the shortening, relation degree, held set, multiplicities, and retained derivative levels. A *singleton kernel* omits one anchor. A *relation batch* is one block-Krylov recovery, while a *relation block* is the verified independent collection retained from one or more batches. A *leaf* is one fully guessed support-and-phase tuple. Multi-position covers and common kernels are defined only when used. Appendix A summarizes the status of every priced ingredient.

2 McEliece and the Binary Hold-out Operator

Let $q = 2^m$ and let $\Gamma_2 \subseteq \mathbb{F}_2^n$ be the proper binary Goppa code defined by a degree-t irreducible Goppa polynomial and a length-n support over $\mathbb{F}_q$. Its dimension is $k = n - mt$. We use the generator-matrix public-key presentation of

McEliece [13]; the Niederreiter formulation used by Classic McEliece is equivalent for key recovery [4]. Public row operations and column permutations put a shortening of length ℓ into systematic form

$$\mathbf{G}_\ell = [I_{k_\ell} \mid R_\ell], \qquad n_\ell = n - \ell, \qquad k_\ell = k - \ell, \qquad n_\ell - k_\ell = mt. \tag{1}$$

Write $p_j \in \mathbb{F}_2^{k_\ell}$ for its columns. The first k_ℓ columns are the weight-one pivots. In Vedenev's parametrization, the non-pivot columns have the hidden form $p_j = \lambda_j F(a_j)$, where $\lambda_j \in \mathbb{F}_q^*$ and $F \in \mathbb{F}_q[Z]^{k_\ell}$ is a coefficient curve of degree at most

$$D_\ell = n_\ell - 2t - 1 = k_\ell + mt - 2t - 1. \tag{2}$$

Let Π_d be the $\mathbb{F}_2$-space of square-free homogeneous degree-d forms in k_ℓ variables. It has the monomial basis $X_S = \prod_{i\in S} X_i$ indexed by $S \in \binom{[k_\ell]}{d}$, and hence dimension

$$N = \dim \Pi_d = \binom{k_\ell}{d}. \tag{3}$$

Choose a set $\mathcal{H}$ of h anchors, meaning held-out non-pivot columns. Each remaining non-pivot column p receives a Hasse vanishing order $s_p \le d-1$. For $T \subseteq [k_\ell]$ with $|T| < s_p$, the corresponding row of the binary hold-out operator M is

$$M[(p,T), S] = [T \subseteq S] \prod_{i \in S\setminus T} p_i = [T \subseteq S][S \setminus T \subseteq \operatorname{supp}(p)]. \tag{4}$$

Thus $M\boldsymbol{\eta} = 0$ says that the selected Hasse derivatives of $H = \sum_S \eta_S X_S$ vanish at every constrained public column. Equation (4) is the common binary incidence operator underlying the HOV and hold-out constructions; the two uses differ in their input columns, degree, retained derivative levels, and held-out set [8,11,15].

3 Exact Relation-Generation Reductions

3.1 Hermite Soundness and Heterogeneous Multiplicity

The systematic pivots contribute vanishing for free. A square-free degree-d monomial needs d distinct nonzero coordinates, so at a weight-one pivot every derivative of order less than $d-1$ vanishes without adding a row to M; this is the systematic-form optimization of [8,15]. Our Hermite accounting additionally carries that free order-$d-1$ multiplicity into the interpolation inequality below.

Lemma 1 (Hermite soundness). *Let $H \in \Pi_d$ vanish to order s_p at each constrained non-pivot column. If*

$$k_\ell(d-1) + \sum_{p \notin \mathcal{H},\, p \text{ nonpivot}} s_p > dD_\ell, \tag{5}$$

then $H \circ F$ is the zero polynomial. In particular, H and all consequences of the identity $H \circ F = 0$ remain valid at every anchor.

Proof. The univariate polynomial $H \circ F$ has degree at most dD_ℓ. Its Hasse zero multiplicities include $d-1$ at each of the k_ℓ systematic pivots and s_p at each constrained non-pivot node. Distinct nodes with total multiplicity greater than the degree force the polynomial to vanish identically.

Put $\delta_H = dD_\ell - k_\ell(d-1)$ and let $v = mt - h$ be the number of constrained non-pivot columns. The smallest integral order sum admitted by Lemma 1 is $\delta_H + 1$. Existing instantiations use one uniform order, which changes this sum in steps of v; a heterogeneous schedule changes it one unit at a time.

For one public column p assigned order s, its *point block* is the collection of operator rows indexed by all T with $|T| < s$. We call this block *stable-weight* when p has Hamming weight $w \geq d+s-1$. In this range its exact rank is $\binom{k_\ell}{s-1}$, as shown by (11) below.

Proposition 1 (Balanced heterogeneous schedule). *Assume* $2 \leq \lceil(\delta_H + 1)/v\rceil \leq d-1$*, that the compared point blocks are in the stable-weight range, and that*

$$d - 2 \leq \frac{k_\ell - \sqrt{k_\ell + 2}}{2}. \tag{6}$$

Define

$$s_{\mathrm{hi}} = \left\lceil \frac{\delta_H + 1}{v} \right\rceil, \qquad a = \delta_H + 1 - v(s_{\mathrm{hi}} - 1). \tag{7}$$

Assign order s_{hi} *to* a *columns and order* $s_{\mathrm{hi}} - 1$ *to the other* $v - a$ *columns. This schedule has Hermite slack one. In the stable-weight range, it minimizes the summed point-block rank among all schedules satisfying* (5)*.*

Proof. The displayed schedule has total order $\delta_H + 1$, hence slack one. In the stable-weight range, the rank charged by order s is $f(s) = \binom{k_\ell}{s-1}$. Writing $j = s-1$, its second difference satisfies

$$\frac{f(s+1) - 2f(s) + f(s-1)}{f(s)} = \frac{(k_\ell - 2j)^2 - k_\ell - 2}{(j+1)(k_\ell - j + 1)}. \tag{8}$$

Condition (6) makes this nonnegative for every $1 \leq j \leq d-2$; it also places these indices below $k_\ell/2$, where f is increasing. Hence any schedule with order sum greater than $\delta_H + 1$ can be reduced to sum $\delta_H + 1$ without increasing its rank charge. If two remaining orders differ by at least two, discrete convexity shows that transferring one unit from the larger to the smaller does not increase their summed charge. Repeating this exchange leaves only the two adjacent orders in (7).

In the tables, $a@s_{\mathrm{hi}} + (v-a)@s_{\mathrm{lo}}$ abbreviates a columns assigned order s_{hi} and $v - a$ columns assigned order s_{lo}.

The proposition is more than a constant-factor optimization. In every cell of Table 2, the largest uniform order whose row-rank bound fits below N is exactly one below the smallest uniform order satisfying Hermite. For those cells, the uniform feasibility window is empty, while the heterogeneous schedule lands between its two endpoints. This statement does not apply to every later Pareto or sensitivity cell; Table 3, for example, displays an admitted uniform schedule at $(k_\ell, d) = (215, 8)$. Mixed multiplicity enlarges the kernel relative to the high uniform order; semantic validity, rather than equality of kernels, follows from Lemma 1.

3.2 Lucas-Minimal Derivative Levels

Let $W_{i,j}$ denote the binary inclusion matrix from i-subsets to j-subsets. Wilson's decomposition of inclusion matrices and Lucas's parity criterion give

$$\begin{gathered} W_{i,j} W_{j,d} = \binom{d-i}{j-i} W_{i,d}, \\ \binom{d-i}{j-i} \equiv 1 \pmod 2 \iff (j-i) \mathbin{\&} \neg(d-i) = 0. \end{gathered} \tag{9}$$

We use the resulting binary row-space statement [16]; related modular-rank formulas appear in [2,7]. Wilson's common diagonal form groups its basis positions into degree-i families, which we denote by $\mathcal{D}_i$.

Theorem 1 (Lucas level compression). *For $0 \le j < s$, define*

$$\mathcal{U}(d, j) = \{i : 0 \le i \le j,\ (j-i) \mathbin{\&} \neg(d-i) = 0\}. \tag{10}$$

If $J \subseteq \{0, \ldots, s-1\}$ satisfies $\bigcup_{j \in J} \mathcal{U}(d, j) = \{0, \ldots, s-1\}$, then the Hasse rows at levels J span the same binary row space as all levels $0, \ldots, s-1$. Consequently, the compression preserves the kernel for a fixed multiplicity schedule.

Proof. The level-j block sees the family $\mathcal{D}_i$ precisely when the coefficient in (9) is odd. The full stack sees every $\mathcal{D}_i$ because level $j = i$ does. The covering condition says that the retained levels see the same families, and Wilson's diagonal form then gives equality of row spaces.

The minimum-cardinality set $J(d, s)$ is found by a small exhaustive set-cover search over the $s \le d$ levels. For $(d, s) = (8, 7)$ and $(8, 6)$ it gives $\{0, 4, 6\}$ and $\{0, 4, 5\}$; for $(9, 8)$ and $(9, 7)$ it gives $\{1, 5, 7\}$ and $\{1, 5, 6\}$.

3.3 Rank, Nonzeros, and Cell Selection

Proposition 2 (Exact point-block rank). *At a binary point of weight w, the order-s point block has rank*

$$\rho_{\mathrm{pt}}(k_\ell, w; d, s) = \sum_{e=0}^{s-1} \binom{k_\ell - w}{e} \binom{w}{\min\{s-1-e, d-e, w-d+e\}}, \tag{11}$$

where out-of-range binomial coefficients are zero. If $w \ge d + s - 1$, this is $\binom{k_\ell}{s-1}$.

Proof. Let $P = \mathrm{supp}(p)$ and $C = [k_\ell]\backslash P$. For a degree-d column S, put $A = S \cap C$ and $e = |A|$. Equation (4) is nonzero only when the outside-support part of the row T is exactly A. After permuting rows and columns, the point block is therefore a direct sum over $A \in \binom{C}{e}$, $e < s$. The summand for A stacks the inclusion matrices $W_{i,d-e}$ on the w coordinates of P for $0 \leq i \leq s-1-e$.

In the common Bier–Wilson diagonal basis [16,2], the degree-j family has multiplicity $\binom{w}{j} - \binom{w}{j-1}$. It occurs in this stack exactly for

$$0 \leq j \leq \min\{s-1-e, d-e, w-d+e\}.$$

Indeed, level $i = j$ is present and has diagonal coefficient $\binom{d-e-j}{0} = 1$; no larger j occurs in the row or column level. Summing the family multiplicities telescopes to the second binomial coefficient in (11), and there are $\binom{k_\ell - w}{e}$ choices of A. In the stated stable-weight range the minimum is $s-1-e$, so Vandermonde's identity gives $\sum_e \binom{k_\ell - w}{e}\binom{w}{s-1-e} = \binom{k_\ell}{s-1}$.

Proposition 2 and the nonzero formula were independently checked with literal binary matrices. The checks cover nine registered generated-data shapes and all 870 valid tuples with $k_\ell \leq 8$. The proof, rather than this bounded experiment, justifies the target-scale tally. The stable value also upper-bounds the low-weight cases, so

$$\overline{\rho} = \sum_p \binom{k_\ell}{s_p - 1} \tag{12}$$

is an upper bound on the global rank of the stacked operator, and

$$\dim \ker M \geq N - \overline{\rho} \tag{13}$$

is a rigorous dimension guarantee whenever the right-hand side is positive. We neither require nor assert cross-point independence here.

For a point of weight w and retained level set J, the exact number of nonzeros is

$$\nu(w; d, J) = \sum_{j \in J} \sum_{e=0}^{j} \binom{w}{e}\binom{k_\ell - w}{j - e}\binom{w - e}{d - j}. \tag{14}$$

The total ν is the sum of (14) over constrained columns with their assigned levels. The model uses the uniform-weight convention $w = \lfloor k_\ell/2 \rfloor$ for every non-pivot column and does not optimize which columns receive the lower order. The nonzero count is therefore modeled and weight-dependent even though the cell dimensions and row-rank bound are exact.

We write

$$R_M = \sum_p \sum_{j \in J_p} \binom{k_\ell}{j} \tag{15}$$

for the number of emitted rows after Lucas compression. The local certificate of Algorithm 4 needs $\mathrm{rank}\, \Psi_1 = k_\ell - (m+1)$ and then $\mathrm{rank}\, \Psi_2 = \binom{m}{2}$ on $\ker \Psi_1$. A

Algorithm 1: SelectCells() : exact admissibility and modeled relation-generation cost.

Input: (n, m, t); finite ranges for (k_ℓ, d, h); a column-weight model.
Output: All admissible cells with schedules, level sets, dimensions, nonzeros, anchor counts, and resource estimates.

```
Initialize an empty list C;
foreach candidate (k_ℓ, d, h) in the specified ranges do
    Set n_ℓ = k_ℓ + mt, D_ℓ = n_ℓ − 2t − 1, v = mt − h, and δ_H = dD_ℓ − k_ℓ(d − 1);
    Compute (s_hi, a, s_lo, v − a) from (7); refuse unless 2 ≤ s_hi ≤ d − 1;
    Find minimum covers J_hi = J(d, s_hi) and J_lo = J(d, s_lo) using (10);
    Set N = binom(k_ℓ, d) and ρ̄ = a·binom(k_ℓ, s_hi − 1) + (v − a)·binom(k_ℓ, s_lo − 1);
    Refuse if ρ̄ ≥ N;
    Sum (14) to obtain ν and count the emitted operator rows R_M;
    Set U = k_ℓ(D_ℓ + 1), B = binom(k_ℓ, 2), c = max{4, ⌈U/B⌉}, and r_min by (16);
    Append the complete record, including the kernel floor N − ρ̄, to C;
end
return C;
```

relation block can support both ranks only if its dimension is at least their sum, so throughout we set

$$r_{\min} = k_\ell - (m+1) + \binom{m}{2}. \tag{16}$$

This gives five width-64 batches at the main mceliece348864 cells used below. It is only the dimension required by the first-order rank checks; it does not imply that the truncated block determines the systems used for higher-order derivative recovery. Assumption 2 isolates that additional interface.

Algorithm 1 combines Proposition 1 and Theorem 1. The implemented scan is exhaustive over the stated (k_ℓ, d, h) ranges after fixing the proved balanced schedule, the lexicographically first minimum-cardinality Lucas cover, and the modeled non-pivot weight. It does not search arbitrary multiplicity schedules or optimize among equal-cardinality covers. Exhaustion over k_ℓ remains necessary because the nonzero count is not monotone in either dimension or multiplicity.

We call a cell *dimension-admitted* when it passes the multiplicity-order condition and Hermite inequality in Algorithm 1 and satisfies $\overline{\rho} < N$. Admission guarantees the positive kernel floor (13); it does not guarantee Krylov yield or downstream recovery.

The selector does not operate on the public key beyond reading its parameter shape. For Q candidate triples, minimum-level set cover takes at most 2^d subset tests per multiplicity, and the rank and nonzero sums take $O(d^3)$ exact-integer operations, giving $O(Q(2^d + d^3))$ small combinatorial operations. This one-time model-construction cost is excluded from the cryptanalytic bit tally. The implemented scan is restricted to $h = 1$ and minimizes the width-64 one-position relation-generation cost; cover and common hold-outs require separate richness assumptions and kernel counts.

4 Matrix-Free Relation Generation

For a retained row (p, T) at level j, matrix application is the restricted $(d-j)$-step shadow

$$(M\boldsymbol{\eta})_{p,T} = \bigoplus_{\substack{V \subseteq \mathrm{supp}(p) \setminus T \\ |V| = d-j}} \eta_{T \cup V}. \tag{17}$$

The tempting square operator $M^{\mathsf{T}}M$ need not have the desired kernel over $\mathbb{F}_2$: if $0 \neq v \in \operatorname{im} M \cap \ker M^{\mathsf{T}}$ and $Mx = v$, then $x \in \ker(M^{\mathsf{T}}M) \setminus \ker M$. A simpler exact squaring is available at essentially every admitted cell.

Lemma 2 (Injectively padded right kernel). *Suppose $R_M \leq N$, and let $E : \mathbb{F}_2^{R_M} \hookrightarrow \mathbb{F}_2^N$ be any injective linear map. For the square matrix*

$$B = EM \in \mathbb{F}_2^{N \times N}, \tag{18}$$

one has $\ker B = \ker M$ *and* $\operatorname{rank} B = \operatorname{rank} M \leq \overline{\rho}$.

Proof. Injectivity gives $EMx = 0$ if and only if $Mx = 0$, and it preserves the dimension of the image of M.

By Lemma 2, zero-padding is a valid E, so one application of B consists of one traversal of M followed by a free coordinate embedding. Its recovered right kernel is already $\ker M$; unlike augmentation, it introduces no left-kernel component that must be projected away. Coppersmith's method applies to an arbitrary square matrix and does not require symmetry [5]. If $R_M > N$, we retain the exact augmented fallback $\mathcal{A} = \left(\begin{smallmatrix} 0 & M^{\mathsf{T}} \\ M & 0 \end{smallmatrix}\right)$, for which projection returns $\ker M$. Put

$$(D_\star, R_\star, \tau) = \begin{cases} (N, \overline{\rho}, 1), & R_M \leq N \quad \text{(padded)}, \\ (N + R_M, 2\overline{\rho}, 2), & R_M > N \quad \text{(augmented)}, \end{cases} \tag{19}$$

where τ counts sparse traversals per solver application. Only 29 of the 19,338 admitted mceliece348864 singleton cells require the fallback, and none is a displayed minimizer.

For one nominal balanced equal-block Coppersmith attempt, choose independent projections $X, Y \in \mathbb{F}_2^{D_\star \times b}$, form the projected sequence, compute a minimal matrix generator, and perform the recovery pass. The retained precision and application count are

$$L_b = 2\left\lceil \frac{R_\star}{b} \right\rceil + b, \qquad n_\star = 3\left\lceil \frac{R_\star}{b} \right\rceil. \tag{20}$$

Let $u = c$ for singleton kernels, $u = \lceil c/h \rceil$ for a cover, and $u = 1$ for one common kernel. Put $n_b = \lceil r_{\min}/b \rceil$ and $w_b = 64\lceil b/64 \rceil$, the bit-operation charge for one packed block XOR per incidence. If r_{BW} attempts are executed per required batch, then

$$C_{\mathrm{rel}}(b) = 3\tau w_b \left\lceil \frac{R_\star}{b} \right\rceil \nu\, u n_b r_{\mathrm{BW}}. \tag{21}$$

Apart from integer ceiling effects, the padded width-64 price is four times smaller than the former augmented price; at every displayed cell the change is -2.000 bits to the shown precision. Widths above 64 use an explicit multiword model rather than pretending that a width-b block costs exactly b bit operations. Choosing the smallest multiple of 64 at least $r_{\min}$ reduces five or six sequential batches to one, but still needs a reliable wide-block implementation and charges the larger state below.

Every displayed row sets $r_{\mathrm{BW}} = 1$. If p_{BW} is the unknown per-attempt failure probability, expected work contains $1/(1 - p_{\mathrm{BW}})$. Coppersmith excludes pathological matrices from the generic projection argument [5]; Eberly's reliable block-Lanczos theorem is over large fields, while his small-field treatment uses sparse preconditioning and partial Frobenius decomposition [6]. A sparse invertible preconditioner or a different injection E can be used on retry, but we give no reliability theorem for these structured binary matrices. Equation (21) is therefore a verified-or-refused one-attempt tally, not an expected running time.

The solver state, retained relations, and active peak are distinct. We retain up to the guaranteed capacity $r_b = \min\{N - \bar{\rho}, b n_b\}$ from successful batches, rather than discarding surplus output, and set

$$\begin{aligned}
M_b^{\mathrm{vec}} &= \frac{bD_\star}{8}, & M_b^{\mathrm{seq}} &= \frac{L_b b^2}{8},\\
M_b^{\mathrm{solve}} &= M_b^{\mathrm{vec}} + M_b^{\mathrm{seq}}, & M_b^{\mathrm{rel}} &= \frac{r_b N}{8},\\
M_b^{\mathrm{active}} &= \max\left\{ M_b^{\mathrm{solve}} + \frac{(n_b - 1)bN}{8}, M_b^{\mathrm{rel}} \right\}.
\end{aligned} \tag{22}$$

The first active term stores previous batches while reducing the last batch against them; the second stores the final certified block. Tables report $\log_2(8M_b^{\mathrm{active}})$. At the displayed $c = 7$ cell, padding lowers solver-only state from $2^{61.660}$ to $2^{60.738}$ bits; including earlier relation batches, the active peak falls from $2^{62.549}$ to $2^{62.124}$ bits. Recursive generator-basis temporaries, checkpoints, random access, I/O, and replication remain excluded.

Keeping surplus output is free at the sparse-application level and gives the local rank tests useful oversampling. Accepted relations may be repacked into $n_{64} = \lceil r_b/64 \rceil$ words for fresh annihilation checks. Together with naive independence testing, this adds at most

$$C_{\mathrm{replay}} \le u\left(64 n_{64}\nu + r_b^2 N\right) \tag{23}$$

bit operations. We record this as a separate lower-order term rather than changing the leading relation-generation exponent; at every displayed cell it is far below C_{rel}.

Algorithm 2: GenerateRelations() : matrix-free certified right-kernel generation.

Input: A cell from Algorithm 1, its constrained public columns, block width b, and retry limit $R_{\max}$.
Output: At least $r_{\min}$ independent relations in $\ker M$, or $\perp$.

Implement ApplyM by the shadow traversal (17); use $\mathsf{ApplyB}(x) = E\,\mathsf{ApplyM}(x)$ when $R_M \le N$, and the augmented fallback otherwise;
Initialize an empty certified relation block K';
for $j = 1$ **to** $n_b = \lceil r_{\min}/b \rceil$ **do**
  Set the required cumulative rank $r_j = \min\{jb, r_{\min}\}$ and mark this batch unfinished;
  for $r_{\mathrm{BW}} = 1$ **to** $R_{\max}$ **do**
    Draw fresh $X, Y \in \mathbb{F}_2^{D_\star \times b}$ and the selected preconditioner;
    Form the projected sequence through precision L_b; compute a minimal generator and run Coppersmith recovery;
    Undo preconditioning, project only for the augmented fallback, verify every vector with ApplyM, and append every new independent relation to K';
    if $\dim K' \ge r_j$ **then**
      Mark the batch finished; **break**;
    end
  end
  if *the batch is unfinished* **then**
    return $\perp$;
  end
end
Replay the Hermite inequality, schedule, Lucas covers, and all operator annihilations; **return** K';

For a held set $\mathcal{H}$, write $M_{\mathcal{H}}$ for the operator with those columns omitted and $K_{\mathcal{H}} = \ker M_{\mathcal{H}}$. At a fixed multiplicity assignment on the common constrained columns, $i \in \mathcal{H}$ implies $K_{\{i\}} \subseteq K_{\mathcal{H}}$ because holding out more columns removes rows. The costed h-position cells are then re-scheduled so that (5) stays strict; no containment between the re-scheduled kernels is asserted. A cover of the c anchors by h-position hold-outs reduces the number of relation-generation computations to $\lceil c/h \rceil$, but the containment fact alone does not say that their certified blocks determine the same nested derivative filtration at each anchor.

Assumption 1 (Multi-position hold-out richness) *For every held set $\mathcal{H}$ in the chosen cover, public restrictions of a certified block from $K_{\mathcal{H}}$ give the same nested derivative filtration at each $i \in \mathcal{H}$ as a sufficiently rich block from $K_{\{i\}}$.*

At the $c = 8$, $k_\ell = 215$, $d = 8$, $b = 32$ cell, every $h \le 6$ is dimension-admitted. The direct relation-generation prices in Table 1 show that two disjoint $h = 4$ kernels dominate the proposed two-kernel $h = 6$ cover: they are slightly cheaper, use less state, avoid overlap, and require the weaker enlargement. Every $h > 1$ row is conditional on Assumption 1.

Table 1. Conditional multi-position hold-out covers of the eight anchors at $(k_\ell, d, b) = (215, 8, 32)$. Work is $\log_2$ bit operations; Δ is relative to eight one-position relation-generation computations.

h	Kernels	Kernel floor	Relation generation	Δ	$\log_2$ state bits
1	8	6,279,785,122,167	114.71	+0.00	54.94
2	4	5,538,193,278,188	113.71	-0.99	54.94
3	3	4,796,601,434,209	113.31	-1.40	54.94
4	2	4,055,009,590,230	112.73	-1.97	54.94
5	2	3,313,417,746,251	112.74	-1.96	54.95
6	2	2,571,825,902,272	112.75	-1.95	54.95

5 From Relations to a Key

At an anchor y, the local construction first separates candidate tangent branches and then performs higher-order derivative recovery by solving sequential systems for an indexed nested filtration [15]. These are distinct interfaces. The saturation order r_i can lie anywhere in $[k_\ell - 1, D_\ell]$, and the filtration may have plateaus. The algorithm and arithmetic model below deliberately price the strict-profile special case $r_i = k_\ell - 1$ with a one-dimensional increase at every order. For a relation space K' let $\Psi_1(H) = \nabla H(y)$, let $T_0 = \ker \Psi_1^{\mathsf{T}}$, and let Ψ_2 be the quadratic Hasse map induced by $\ker \Psi_1$ on $T_0/\langle y \rangle$, written in the degree-two coordinate space of that quotient. A *branch plane* is the two-dimensional space $\langle y, v_1 \rangle$ determined by one candidate first-order tangent direction v_1.

Put $V = T_0/\langle y \rangle$ and $\mathcal{Q} = \operatorname{im} \Psi_2 \subseteq \mathbb{F}_q[V]_2 = \operatorname{Sym}^2(V^*)$. The following bounded multiplication-matrix routine makes branch extraction explicit. Its trial forms come from a public reproducible uniform stream, and exhausting the limit R_{br} refuses the block rather than guessing a branch.

These checks certify the complete first-order branch scheme of an accepted block. Indeed, use the m spanning quotient lines as coordinate axes. Exact vanishing removes every square coefficient from the public quadrics, so their image lies in the $\binom{m}{2}$-dimensional cross-term space. The checked rank forces equality, and the common zero scheme of all cross terms is exactly the reduced union of those axes. For a genuine hold-out relation block, the true tangent orbit lies in this scheme; it is a nonempty Frobenius-stable subset of the checked single m-cycle and hence equals the whole scheme. Thus an accepted genuine block does not additionally rely on Vedenev's Conjecture 3 for branch separation; higher continuation remains conditional.

A representation-independent certification envelope follows by scanning every degree-d coefficient once for the value, gradient, and quadratic Hasse maps. Put $s_2 = \binom{m+1}{2}$. Naive quotient reduction, multiplication matrices, the deterministic

Algorithm 3: ExtractPureBranches() : fail-closed quadratic-scheme decomposition.

Input: V, a basis of $\mathcal{Q} \subseteq \mathrm{Sym}^2(V^*)$, and trial limit R_{br}.
Output: m projective quotient lines, or $\perp$.

1 Row-reduce $\mathcal{Q}$ and refuse unless $\dim \mathcal{Q} = \binom{m}{2}$;
2 Form $A_2 = \mathrm{Sym}^2(V^*)/\mathcal{Q}$ and, for a basis $x_1, \ldots, x_m$ of V^*, the maps $\mu_x : V^* \to A_2$, $f \mapsto xf$;
3 Try at most R_{br} stream forms ℓ until μ_ℓ is invertible; otherwise **return** $\perp$;
4 Put $M_{x_i} = \mu_\ell^{-1}\mu_{x_i}$ and extend $x \mapsto M_x$ linearly on V^*;
5 For each of at most R_{br} stream forms h, scan every $\lambda \in \mathbb{F}_q$ and row-reduce $M_h^{\mathsf{T}} - \lambda I$; accept the first h for which the nonzero kernels occur at exactly m values, are all one-dimensional, and have direct sum of dimension m;
6 If no h is accepted, **return** $\perp$; otherwise refuse unless its m left eigenspaces are common left eigenspaces of every M_{x_i};
7 Interpret them in $(V^*)^* = V$; refuse unless the resulting lines are distinct, span V, vanish on every element of $\mathcal{Q}$, and form one coefficientwise-squaring cycle;
8 **return** *the m quotient lines*;

Algorithm 4: CertifyFirstOrder() : first-order branch certification.

Input: A certified relation block $K' \subseteq \ker M$, anchor y, extension degree m, and limit R_{br}.
Output: The m candidate branch planes through y, or $\perp$.

1 Check $H(y) = 0$ for every $H \in K'$; otherwise **return** $\perp$;
2 Construct Ψ_1, put $T_0 = \ker \Psi_1^{\mathsf{T}}$, and check $\dim T_0 = m + 1$ and $y \in T_0$;
3 Put $K_0' = \ker \Psi_1$, construct $\Psi_2 : K_0' \to \mathbb{F}_q[T_0/\langle y\rangle]_2$, and check $\operatorname{rank} \Psi_2 = \binom{m}{2}$;
4 Call ExtractPureBranches on $V = T_0/\langle y\rangle$ and $\operatorname{im} \Psi_2$; refuse on $\perp$;
5 Lift its quotient lines with $\langle y\rangle$ and replay exact vanishing, distinctness, spanning, and the coefficientwise-squaring m-cycle;
6 **return** *the certified branch planes, otherwise* $\perp$;

eigenvalue scan, and eigenspaces give

$$\begin{aligned} C_{\mathrm{br}} &= O\big((s_2^3 + R_{\mathrm{br}} q m^3) m^2\big), \\ C_{\mathrm{cert},1} &\le r_{\min} N \left(1 + d + \binom{d}{2}\right) + O\left(r_{\min} k_\ell^2 + r_{\min} \binom{m}{2}^2\right) + C_{\mathrm{br}}, \qquad (24) \\ C_{\mathrm{cert}} &= c\, C_{\mathrm{cert},1}. \end{aligned}$$

The first two terms are binary operations; C_{br} uses the same m^2 bit-operation proxy as the extension-field arithmetic below. The artifact instantiates the displayed O-terms with fixed reproducible constants, so its certification number is an accounting envelope rather than an implementation-resolved upper bound. For a reduced spanning m-point scheme, evaluation identifies V^* and A_2 with $\mathbb{F}_q^m$: μ_ℓ is invertible exactly when ℓ misses every point, and the common left eigenvectors are their evaluation functionals. Each trial form is uniform; a de-

Algorithm 5: $\mathsf{RecoverHigherDerivatives}()$: higher-order derivative recovery at one anchor.

Input: K', anchor y, one certified first-order plane P, and branch limit r_{der}.
Output: A complete nested derivative filtration, or $\perp$.

```
Set v_0 = y, choose v_1 ∈ P \ ⟨y⟩, and set W^[j] = ⟨v_0, ..., v_j⟩ for j = 0, 1;
for r = 2 to k_ℓ − 1 do
    Construct both affine systems in (25) from K' and the retained partial branches;
    Solve them over F_{2^m}; substitute every solution into every equation and discard inconsistent branches;
    Quotient solutions by the current W^[r−1], retain at most r_der distinct extensions, and refuse on overflow;
    For each retained extension, set W^[r] = W^[r−1] + ⟨v_r⟩ and require dim W^[r] = r + 1;
end
Refuse unless the surviving nested spaces form the unique complete filtration associated with P;
return that filtration;
```

nominator trial fails with probability at most m/q, and a separator collides with probability at most $\binom{m}{2}/q$. The fixed retry cap can therefore refuse a genuine block, but exact replay prevents false acceptance. With $R_{\mathrm{br}} = 64$ and $m \leq 13$, this decomposition is lower-order in every displayed cell.

For $H \in K'$ and vectors $v_1, \ldots, v_{r-1}$, let $\Phi_{H,r}(y; v_1, \ldots, v_{r-1})$ denote the coefficient of ϵ^r in the terms of Hasse order at least two in $H(y + \sum_{j<r} v_j \epsilon^j)$. The sequential equations used for higher-order derivative recovery in [15] determine the next vector v_r from

$$\begin{aligned} \nabla H(y) \cdot v_r &= -\Phi_{H,r}(y; v_1, \ldots, v_{r-1}) \quad &&(H \in K'), \\ \Phi_{H,r+1}(y; v_1, \ldots, v_r) &= 0 &&(H \in \widehat{K}'), \end{aligned} \tag{25}$$

where $\widehat{K}' = \{H \in K' : \nabla H(y) = 0\}$. The equations are affine in v_r, but their consistency and semantic completeness are not consequences of the first-order rank check.

Assumption 2 (Relation-block richness) *The binary-Goppa relation spaces used here satisfy Vedenev's Conjecture 4 [15]. Moreover, the recovered filtration has the strict profile* $\dim V_i^{[r]} = r + 1$ *through saturation at* $r_i = k_\ell - 1$*, and the certified* $r_{\min}$*-dimensional truncated block* K' *is rich enough that every sequential system* (25) *has the same admissible branches as the full hold-out kernel. The total number of retained affine branches is at most* r_{der}.

For completeness of the strict-profile subtotal, we charge a conservative dense implementation rather than calling this stage free. Rescanning every coefficient to assemble all truncated Hasse expressions and densely solving every k_ℓ-variable system gives the one-time upper model

$$\begin{aligned} C_{\mathrm{der}} &\le cmr_{\mathrm{der}}(C_{\mathrm{asm}} + C_{\mathrm{elim}})m^2, \\ C_{\mathrm{asm}} &= r_{\min} N d k_\ell^3, \\ C_{\mathrm{elim}} &= (k_\ell - 2)(r_{\min} k_\ell^2 + k_\ell^3), \end{aligned} \tag{26}$$

bit operations. The initial factor m charges all certified first-order branches at each anchor, and the final m^2 is the same optimistic field-multiplication proxy used below. The tables set the branch limit $r_{\mathrm{der}} = 1$. Equation (26) prices arithmetic and possible branch retries multiplicatively; it does not prove Assumption 2. In the displayed cells, it is lower order than relation generation or candidate interpolation.

Suppose Algorithm 5 yields strict complete derivative filtrations at c anchors, with basis matrices $Q_i \in \mathrm{GL}_{k_\ell}(\mathbb{F}_q)$ and candidate support coordinates $a_i \in \mathbb{P}^1(\mathbb{F}_q)$. The *candidate-interpolation solve*, corresponding to Step 3 of [15], seeks a polynomial vector in the intersection

$$\mathcal{L} = \bigcap_{i=1}^{c} Q_i \operatorname{diag}\bigl(1, Z - a_i, \ldots, (Z - a_i)^{k_\ell - 1}\bigr)\mathbb{F}_q[Z]^{k_\ell} \tag{27}$$

with degree at most D_ℓ. It has

$$\begin{array}{ll} U = k_\ell(D_\ell + 1) & \text{coefficient unknowns,} \\ B = \binom{k_\ell}{2} & \text{strict-profile conditions per anchor,} \\ c = \max\{4, \lceil U/B \rceil\} & \text{anchors.} \end{array} \tag{28}$$

We use c to reject an *incorrect* guess under Conjecture 1 of [15]; this does not imply that the correct Goppa leaf is one-dimensional. The lower bound $c \ge 4$ accounts for the residual projective gauge, while $B = \binom{k_\ell}{2}$ is the strict-profile count of [15, Remark 6]. This lower bound is active in 4,592 of the 19,338 admitted one-position cells scanned for mceliece348864, although none of the displayed cells has fewer than five anchors. The selector retains cells for which the nested prefix model is unavailable, but the nested price is refused whenever $U - (c-1)B < 2$.

Apon proves an algebraic obstruction on a *correct* binary-Goppa leaf [1]. If the hidden coefficient curve F and its formal derivative F' are not proportional over $\mathbb{F}_q(Z)$, then for $c \le 2t + 2$ correctly labelled distinct anchors,

$$\dim_{\mathbb{F}_q} \mathcal{L}_{\le D_\ell} \ge 2t + 4 - c. \tag{29}$$

The binary identity $\varPi F' + \varPi' F = \kappa G^2 F^{(2)}$, with coordinatewise squaring on the right, constructs $2t+3$ directions modulo the true curve: at a correct anchor a, all conditions in its derivative filtration depend on the direction polynomial A only through $A(a)$. The strict first-order profile in Assumption 2 makes $F(a)$ and $F'(a)$ independent, so it implies the stated nondegeneracy. The order-zero part of the same argument applies to a later public value-line restriction. After e further distinct correct labels, the protected subspace therefore still has dimension at least $2t+4-c-e$ while $c+e \le 2t+2$; reaching a line requires

$$e \ge 2t+3-c. \tag{30}$$

Thus every selected value of c is a wrong-guess rejection count, not a correct-leaf uniqueness count. Apon's $> 2^{1500}$ estimate correctly prices the Step 3 procedure analyzed there, in which all $c-2$ support values not fixed by normalization are guessed. He also identifies testing candidates against the full public code as an alternative and describes the resulting task as a nontrivial list problem. The contemporaneous 26 August draft by Vedenev [15] instantiates that alternative by matching unused public columns. It reports $c^\star = 2t+3$ empirically for binary Goppa codes; [1, Eq. (9)] observes the same threshold experimentally, while Apon's theorem proves the corresponding lower bound. Assumption 4 makes success on that list problem explicit and also requires a compatible affine chart; we price a fail-closed attempt but prove neither step.

Proposition 3 (Strict-filtration information count). *If the public datum at one anchor is only an unindexed strict complete derivative filtration in a k_ℓ-dimensional coefficient space, then the interpolation-lattice formulation supplies*

$$\sum_{j=0}^{k_\ell-2} (k_\ell - 1 - j) = \binom{k_\ell}{2} = B \tag{31}$$

independent linear incidence conditions at that anchor. Merely computing another derivative after this filtration has saturated cannot replace B by $\binom{k_\ell}{3}$.

Proof. At derivative order j, membership in the $(j+1)$-dimensional filtration subspace has codimension $k_\ell - 1 - j$. Summing these codimensions gives (31), equivalently the dimension of the complete flag variety. Changes of local parameter act lower-triangularly on the jet sequence and preserve only these nested spans. Additional conditions therefore require calibrated jets, transition data, or another invariant beyond the filtration itself.

This rules out the tempting shortcut of assigning $\binom{k_\ell}{3}$ conditions to an "order-three relation" without identifying new public information. It does not discard the derivative-order indices retained by the filtration. If the filtration saturates at r_i and has dimensions $\dim V_i^{[j]}$, its exact indexed condition count is

$$E_i = \sum_{j<r_i} \left(k_\ell - \dim V_i^{[j]}\right) \ge B. \tag{32}$$

Using $E_i > B$ to reduce the number of anchors would require a new wrong-guess rank-independence statement for the measured indexed types. The displayed model deliberately forgets those indices and retains the strict-profile value B. Compare Vedenev's Remark 6 [15].

There are three relevant solve models. Fresh dense elimination costs approximately U^3 field operations. A depth-first prefix tree shares the first $c-1$ intersections; if every prefix has the generic dimension, its most expensive intersection costs approximately

$$C_{\text{nest}} = B\big(U-(c-1)B\big)^2 \tag{33}$$

field operations. This conditional variant requires $U-(c-1)B \geq 2$; cells failing that test remain available for dense or order-basis pricing. The lattice membership in (27) is exactly the simultaneous modular system

$$(Q_i^{-1}G)_j \equiv 0 \pmod{(Z-a_i)^j}, \qquad 1 \leq i \leq c, \quad 1 \leq j < k_\ell. \tag{34}$$

Its total modulus degree is $\sigma = c\binom{k_\ell}{2} = cB$, making shifted minimal approximant or interpolation bases the natural exact representation. Standard order-basis scaling suggests the leading term

$$\widetilde{O}\big(k_\ell^{\omega-1}\sigma\big), \qquad \sigma = cB, \tag{35}$$

but we have not proved that a particular shifted approximant-basis algorithm attains this bound for (34), including output-degree control and constants. Giorgi and Lebreton use order bases inside block Wiedemann [9]; that result alone does not supply the required reduction. Every order-basis row below is therefore an illustrative leading-monomial sensitivity with the explicitly fixed classical exponent $\omega = 3$, not a supported attack implementation.

Proposition 4 (Bundle criterion). *View (27) as a rank-k_ℓ bundle on $\mathbb{P}^1$ with splitting $\mathcal{L} \simeq \bigoplus_j \mathcal{O}(-e_j)$. Then*

$$\sum_j e_j = cB, \qquad \dim \mathcal{L}_{\leq D_\ell} = \sum_j \max(D_\ell - e_j + 1, 0). \tag{36}$$

In particular, if one $e_j = D_\ell$ and all remaining $e_j \geq D_\ell + 1$, then the correct solution space is one-dimensional.

Proof. Each anchor contributes determinant degree B, giving the first identity. The second is the dimension of the degree-bounded sections of the split line bundles on $\mathbb{P}^1$. The final assertion follows term by term.

Assumption 3 (Priced wrong-prefix profile) *Apart from a subexponential number of exceptional support-and-phase prefixes, semantically wrong prefixes of strict derivative filtrations have the generic intersection dimensions used in* (33).

This assumption supplies only the dimensions used by the nested PGL sensitivity. It says nothing about the correct prefix, and (29) rules out the one-dimensional splitting profile in Proposition 4 for every selected correct binary-Goppa leaf. No target-scale reduced lattice basis or prefix-dimension trace has been computed.

The affine enumeration fixes two anchors and guesses $c-2$ support coordinates and $c-1$ relative Frobenius phases [15]. With $q = 2^m$, its guess count is

$$G_{\mathrm{aff}} = q^{c-2}m^{c-1}, \qquad \log_2 G_{\mathrm{aff}} = m(c-2) + (c-1)\log_2 m. \tag{37}$$

Below, G denotes the traversal count under the selected normalization and phase policy, so $G \le G_{\mathrm{aff}}$; equality holds for affine normalization without phase synchronization. Generalized Reed–Solomon (GRS) covariance under $\sigma(z) = (\lambda z + \mu)/(\gamma z + \delta)$ permits three projective anchors to be normalized, reducing the support factor to q^{c-3}. This is the action of $\mathrm{PGL}_2(\mathbb{F}_q)$, abbreviated PGL below. Indeed, homogenizing a degree-$< k'$ polynomial gives

$$\mathrm{GRS}_{k'}\big(\boldsymbol{\alpha}, \boldsymbol{\beta}\big) = \mathrm{GRS}_{k'}\big(\sigma(\boldsymbol{\alpha}), \boldsymbol{\beta}(\gamma\boldsymbol{\alpha} + \delta)^{k'-1}\big). \tag{38}$$

Proposition 5 (Finite-anchor PGL normalization). *Three distinct anchors may be normalized to $0, 1, \zeta$ with finite $\zeta \notin \{0,1\}$. At every selected anchor that remains finite, the transformed coefficient curve has the same indexed derivative filtration. Provided $c \le q$, as in every selected cell, among a fixed list of $c-2$ distinct choices of ζ, at least one keeps all c selected anchors finite.*

Proof. Homogenize the degree-D_ℓ coefficient curve and apply the inverse linear change of its two source variables, multiplying by the corresponding degree-D_ℓ linear form. At a finite point away from the induced pole, this multiplies the local curve by a nonzero power-series unit and changes the local parameter with nonzero linear coefficient. Its Hasse jets are therefore related by an invertible triangular matrix at every order, preserving their nested spans. For fixed first, second, and third anchors, varying ζ varies the preimage of infinity bijectively over the source points other than those three. Each of the remaining $c-3$ anchors forbids at most one value of ζ, so $c-2$ choices suffice.

Thus the factor-q support saving does not require a selected anchor at infinity; the bounded fallback among ζ values is lower order and is omitted from the exponent. Equation (38) proves ambient-code covariance, while the proposition handles the derivative filtrations used by candidate interpolation. A complete projective support must still be pole-normalized before the affine Goppa-completion endpoint, and Frobenius phases are enumerated unless the separate synchronization assumption is invoked.

After fixing only 0 and 1, a residual projective gauge remains:

$$\phi_\lambda(z) = \frac{\lambda z}{(\lambda - 1)z + 1}, \qquad \lambda \in \mathbb{F}_q^*. \tag{39}$$

These $q-1$ transformations are exactly the pointwise stabilizer of the two normalized anchors. By (38), they transport a correct nonzero leaf to gauge-equivalent nonzero leaves, which represent equivalent GRS keys rather than wrong guesses. Normalizing a third anchor quotients this action; in the affine model a complete traversal must instead accept any gauge copy that passes the full verifier. Accordingly, "wrong leaf" below always means semantically wrong modulo (39).

5.1 Heuristic List-Problem Continuation, Completion, and Exact Verification

By Eqs. (29) and (30), every selected correct leaf enters with a multidimensional candidate space W and needs many further correct labels; the jointly optimal Category 1 cell has $c = 6$, $\dim W \geq 126$, and $e \geq 125$. Algorithm 7 scans unused public columns and projective field points, retaining a value-line restriction only when one unused point uniquely maximizes the surviving dimension. This is our concrete attempt at Apon's nontrivial public-code list problem; we call it *public-column continuation*, and it does not guess further held support points.

Assumption 4 (Public-column continuation success) *For a correct leaf, Algorithm 7 uses unique correct maximizers to reduce W to the true curve line. Then Algorithm 8 reaches a compatible affine chart.*

In experiments over $\mathbb{F}_{2^6}$ supplied with the true initial anchors and derivative filtrations, every accepted step had a unique correct unused maximizer, and fixed-order runs reached dimension one after 9, 12, and 15 total labels for $t = 3, 4, 5$. The latter two exceed $2t + 3$, consistent with Apon's explicit caution that his lower bound is not an equality theorem. A correct value-line restriction can remove up to $k_\ell - 1$ dimensions from the full W, but at most one from Apon's explicit protected subspace. Thus (30) proves a label floor, not a unit-step dimension chain for the whole space or that the floor suffices.

Once $W = \langle G^\star \rangle$, evaluate $G^\star$ throughout $\mathbb{P}^1(\mathbb{F}_q)$, match its projective values to public columns, choose a pole outside the matched support, and convert the labels to a compatible affine chart. Kirshanova–May reconstruction can then recover the Goppa polynomial and extend the remaining support [12]; exact public-key equality alone authorizes output.

Our generated-data endpoint experiment used the 768 systematic labels and the least non-pivot label of a 982-coordinate `mceliece348864` chart. On these 769 correctly labelled, pole-normalized affine points, Kirshanova–May reconstruction returned one irreducible degree-64 factor of multiplicity two, assigned unique labels to all 2,719 omitted coordinates, and passed eleven independent full-key checks. Deterministic producer and replay runs took 365.88 and 370.25 seconds and peaked at 1,526,132 and 1,526,552 KiB. The same labels in the raw projective chart failed global extension immediately. Thus 769 compatible affine labels are sufficient in this experiment, but the pole was supplied by secret-conditioned calibration; the result neither constructs the public derivative filtrations nor proves that a target-scale candidate reaches the required gauge.

Wrong leaves are handled by an explicit heuristic rather than by charging completion at every enumerated tuple.

Assumption 5 (Generic wrong-leaf rank) *Modulo the residual action* (39)*, the $cB \times U$ candidate-interpolation matrix of a semantically wrong support or phase tuple has the nonzero-kernel probability of an independent uniform matrix up to a subexponential factor.*

This assumption is consistent with a bounded $m = 6$ census built from the true secret data: all 5,748 semantic wrong profiles had full rank, while the 63 nonzero profiles were exactly the residual PGL orbit. It is evidence, not a distribution theorem or a target-scale measurement. Its quantitative consequence is evaluated in Section 7.3.

For a candidate key, multiplying its reconstructed $mt \times n$ parity check by the transpose of the public $k \times n$ generator and separately checking rank costs at most

$$C^{\mathrm{up}}_{\mathrm{keycheck}} = 2mtnk + 2(mt)^2 n \tag{40}$$

bit operations. For mceliece348864, the leading product is $2^{33.8}$ bit operations. Appendix B gives a fail-closed implementation and cost envelope for the mandatory correct-leaf continuation.

Algorithm 6 gives the compact outer traversal.

Algorithm 6: RecoverGoppaKey() : conditional search and verification skeleton.

Input: A public McEliece generator $\mathbf{G}'$, parameters (n, m, t), a selected cell, and optimization options.

Output: An equivalent Goppa key, or $\perp$.

Put the selected shortening in systematic form and choose c public anchors;

For each one-position hold-out, call GenerateRelations and CertifyFirstOrder; for every returned plane call RecoverHigherDerivatives; conditionally on Assumption 1, use an admissible multi-position hold-out cover instead;

Normalize two anchors affinely; if the projective option is enabled, choose a finite ζ as in Proposition 5 and normalize three anchors to $0, 1, \zeta$;

Enumerate the remaining support values and relative Frobenius phases; the optional synchronization heuristic is described in Appendix C;

Traverse the remaining support guesses depth first, sharing every prefix lattice intersection;

At each leaf, solve (27); continue if its degree-D_ℓ solution space is zero;

Pass every nonzero solution space to ExtendLeaf; **return** *the first exactly verified key it returns*;

return $\perp$ *if the traversal is exhausted*;

For the actual traversal count $G \leq G_{\mathrm{aff}}$, the accounted downstream tally through candidate interpolation is $Gm^2 C_{\mathrm{solve}}$ in the chosen model. Under Assumption 5, only the correct equivalence orbit and a negligible expected number of wrong leaves enter continuation. The artifact's conservative C_{ext} charges one complete rank-maximization, restoration, completion, and verification attempt; its basis-update allowance also absorbs a fresh dense solve for the exceptional correct prefix. At the displayed Category 1 cells $\log_2 C_{\mathrm{ext}} < 90.6$, so adding one successful correct-orbit attempt changes none of the reported exponents at two decimals. The end-to-end reading nevertheless requires Assumption 4. Appendix B also records the no-success ceiling $C_{\mathrm{cont}} \le GC_{\mathrm{ext}}$; Appendix C records the separate phase-synchronization sensitivity.

6 Cost Model

We express computational work in elementary bit operations. Following NIST's PQC evaluation criteria, Categories 1, 3, and 5 are compared with 2^{143}, 2^{207}, and 2^{272} bit operations, respectively; 2^{128} is a number of AES-128 key checks, not the corresponding operation count.* A width-b sparse block XOR is charged as $w_b = 64\lceil b/64\rceil$ bit operations. One $\mathbb{F}_{2^m}$ multiplication is represented by the optimistic m^2 binary-operation proxy; multiplier XORs are omitted. As a sensitivity range, a 250–300-bit-operation multiplication at $m = 12$ would add 0.80–1.06 bits to downstream exponents, but not to binary relation generation. No field basis, multiplier circuit, or citation resolves that range. These are comparison units, not hardware measurements; random memory access, address generation, communication, and storage are uncharged.

For the dense model only, ordinary row-echelon elimination gives a more resolved sensitivity. With $R = cB \ge U$, the below-pivot updates use

$$M_{\mathrm{GE}} = (R-U)\frac{U(U-1)}{2} + \frac{U(U-1)(2U-1)}{6} \tag{41}$$

field multiplications and the same number of field additions, up to lower-order pivot normalization and inversion. Charging a multiplication at $g_\times$ bit operations and an addition at m bit operations therefore replaces m^2U^3 by $(g_\times + m)M_{\mathrm{GE}}$. At $c = 7$, $m = 12$, and under the assumed $g_\times = 250$–300 range, the accounted subtotal becomes $2^{141.43}$–$2^{141.68}$ bit operations, 0.47–0.72 bits below the deliberately coarse $2^{142.15}$ envelope; its margin below the Category 1 threshold is then 1.32–1.57 bits. The row-echelon operation count is resolved, but the field-multiplier realization is not. This correction does not transfer to (33), whose stated leading monomial is already multiply-shaped.

For a selected solve model, let C_{solve} be one of U^3, $B(U-(c-1)B)^2$, or the leading monomial $k_\ell^{\omega-1}cB$, measured in $\mathbb{F}_q$ operations. Let $f = 2$ for affine normalization and $f = 3$ for the conditional projective normalization. Without phase synchronization, the downstream work is modeled as

$$C_{\mathrm{down}} = 2^{m(c-f)} m^{c-1}\, m^2 C_{\mathrm{solve}}. \tag{42}$$

The replay, all-anchor certification, and one-time strict-profile arithmetic for higher-order derivative recovery are charged by (23), (24), and (26). The main accounted subtotal is

$$C_{\mathrm{acct}} = C_{\mathrm{rel}} + C_{\mathrm{replay}} + C_{\mathrm{cert}} + C_{\mathrm{der}} + C_{\mathrm{down}}. \tag{43}$$

* https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization/evaluation-criteria/security-(evaluation-criteria)

We call affine normalization with the fresh dense solve *Affine+dense*, and three-anchor PGL normalization with the nested prefix solve *PGL+nested*. The displayed end-to-end reading additionally invokes Assumptions 4 and 5. Adding one C_{ext} continuation attempt and its exact key check changes none of the tabulated exponents at two decimals, but continuation success is a necessary heuristic, not an optional fallback. The no-success ceiling C_{cont} in Appendix B is reported separately. Solver retries, access costs, and the other listed omissions remain outside the subtotal. Appendix C gives separate rows that delete the phase factor under an additional unproved pairing assumption and either omit or charge its proposed relation source. Every row uses one nominal binary Coppersmith attempt per relation batch and Assumption 2; Section A collects the status of all ingredients.

The optimization variables of the accounted model are

$$(d, k_\ell, h, \{s_p\}, \{J_p\}, c, b, \text{held-set use}, \text{normalization}, \text{solve model}). \tag{44}$$

The all-target scan below varies every admissible singleton cell and compares $b = 64$ with the smallest multiple of 64 that requests all required relations in one batch. Multi-position covers and the speculative phase synchronizer are reported separately because each needs an additional structural assumption.

7 Concrete Tally

7.1 Selected Relation-Generation Cells

Table 2 gives the heterogeneous cells that minimize modeled relation-generation cost in an exhaustive $h = 1$ search over $3 \leq d \leq 24$ and every admissible k_ℓ, at width 64 and modeled non-pivot weight $\lfloor k_\ell/2 \rfloor$. The displayed cost includes one nominal Coppersmith attempt for each required relation batch on the exact padded operator and all one-position hold-outs. It does not include later work or an expected retry multiplier. All five rows have $d = 8$, high/low Lucas sets $\{0, 4, 6\}/\{0, 4, 5\}$, and Hermite slack one.

Table 2. Selected relation-generation cells for the five Classic McEliece parameter sets. Work columns are base-two logarithms of bit operations with $r_{\text{BW}} = 1$; state is the active retained state M_b^{active} from (22).

Parameter Set	k_ℓ	Schedule	c	$\log_2 N$	$\log_2 C_{\text{rel}}$	$\log_2$ state bits
mceliece348864	208	$719@7 + 48@6$	9	46.11	112.35	54.91
mceliece460896	269	$1228@7 + 19@6$	10	49.12	119.49	58.12
mceliece6688128	305	$1584@7 + 79@6$	12	50.59	123.11	59.59
mceliece6960119	295	$1484@7 + 62@6$	11	50.20	122.10	59.20
mceliece8192128	305	$1584@7 + 79@6$	12	50.59	123.11	59.59

Published hold-out baseline. In the parameter-set order of Table 2, Ghoshal et al. use uniform order $s = 7$ and report [8]

$$\begin{aligned} k_\ell &= (217, 274, 315, 304, 315), \\ \log_2 T_{\mathrm{atk}} &= (114.16, 120.29, 123.95, 123.02, 123.95), \\ \log_2 S_{\mathrm{atk}} &= (66.45, 69.54, 71.37, 70.91, 71.37). \end{aligned}$$

These numbers price a different output: T_{atk} is one reliable small-field block-Lanczos consistency test $M_\tau x = b_\tau$ at one anchor, whereas C_{rel} asks the padded solver for $r_{\min}$ verified right-kernel relations at each of c anchors. For Category 1, five width-64 batches at nine anchors multiply the one-anchor, one-batch charge by 45, and $112.35334 - \log_2 45 \approx 106.86149$. Thus even the apparent proximity of 112.35 and 114.16 compares different outputs and solver models. The storage-exponent gap, $66.45 - 54.91 = 11.54$ bits, is likewise not a measured improvement: their S_{atk} includes the sparse system and conditioned solver state, whereas our active state includes certified relations but excludes conditioning, I/O, and generator-basis temporaries.

The schedules are not cosmetic: every corresponding uniform schedule is refused. The lower adjacent uniform order misses Hermite, while the higher order has a row-rank bound exceeding N. The heterogeneous schedules occupy the intervening one-unit window. The artifact model reproduces the exact search and every row of Table 2.

Hermite slack one is an exact algebraic threshold, not a probabilistic success margin. It is nevertheless useful to ask whether the selected cell depends numerically on equality at one unit. As Table 3 shows, assigning order seven to every constrained column at the $c = 8$, $k_\ell = 215$, $d = 8$ cell buys the maximum admissible slack 42 for only 0.078 bits in the relation-generation exponent; slack 43 would require order eight and is refused by the $s_{\mathrm{hi}} \le d - 1$ bound. Thus the one-unit choice is immaterial to the cost estimate, although additional slack does not strengthen a correctly applied Lemma 1.

Table 3. Hermite-slack sensitivity at $(k_\ell, d, h, b) = (215, 8, 1, 64)$. Work is $\log_2$ relation-generation bit operations.

Slack	Schedule	Kernel floor	Relation generation	Δ
1	$726@7 + 41@6$	6,279,785,122,167	112.86	+0.000
42	$767@7 + 0@6$	1,187,277,582,725	112.93	+0.078

7.2 All Classic McEliece Sets

Table 4 jointly minimizes the *Affine+dense* subtotal over every admitted singleton cell with $3 \le d \le 24$. It includes relation generation, replay, all-anchor certification, the one-time envelope for higher-order derivative recovery, support and phase enumeration, and a fresh U^3 candidate-interpolation solve per guess. The fixed scan uses $b = 64$; the wide sensitivity uses the smallest multiple of 64 that requests $r_{\min}$ relations in one batch. These subtotals stop before continuation; the artifact separately shows that adding one C_{ext} does not change their two-decimal values.

Table 4. Jointly optimized Affine+dense subtotals. A cell is (k_ℓ, d, c); work and active state are base-two logarithms of bit operations and bits. The wide column is a multiword-block sensitivity, not a solver-performance theorem.

Parameter Set	PQ Cat / target	Cell	$b = 64$ Work	$b = 64$ State	One relation batch b	One relation batch Work	One relation batch State
mceliece348864	1 / 143	$(322, 9, 6)$	132.22	65.16	384	129.94	65.95
mceliece460896	3 / 207	$(424, 9, 7)$	152.38	69.13	512	152.38	69.95
mceliece6688128	5 / 272	$(565, 9, 7)$	154.89	73.14	640	154.87	73.84
mceliece6960119	5 / 272	$(525, 9, 7)$	154.24	72.19	640	154.24	72.92
mceliece8192128	5 / 272	$(565, 9, 7)$	154.89	73.14	640	154.87	73.84

All five modeled subtotals fall below their category thresholds, but this comparison is conditional: the binary solver yield, higher-order derivative recovery, generic wrong-leaf behavior, correct-leaf continuation, and passage to compatible affine labels are not proved at target scale. The scan deliberately imposes no memory cap; its fixed-width optima require between $2^{65.16}$ and $2^{73.14}$ active state bits. These are arithmetic optima, not deployment estimates. Their work exponents exceed their active-state exponents by 67.06–83.25 bits, and all five active-state exponents lie within 1.8 bits of the corresponding S_{atk} quoted above; the state definitions differ as already noted. The wide-block improvement is material only for Category 1 because candidate interpolation dominates the other rows. For comparison, jointly optimizing the more conjectural PGL+nested model on mceliece348864 gives $(258, 9, 7)$ at 126.84 for $b = 64$ and 124.67 for $b = 320$.

7.3 The Category 1 Pareto Surface

For mceliece348864, changing the cell moves the anchor count and dominates the downstream. The relevant ladder is shown in Table 5; its width-64 state grows much faster than its arithmetic falls.

Table 5. mceliece348864 anchor ladder at width 64 with $r_{\mathrm{BW}} = 1$ and one-position relation generation. Here $\Delta_{\mathrm{wr}} = cB - U$ is the deterministic row surplus of a wrong-system matrix. *Affine+dense* and *PGL+nested* are defined above.

Anchors c	Config (k_ℓ, d)	Relation generation	Affine guesses ($\log_2$)	Δ_{wr}	Affine + dense	PGL + nested	State $\log_2$ bits
9	$(208, 8)$	112.35	112.68	17,368	172.13	146.29	54.91
8	$(215, 8)$	112.86	97.09	215	156.73	135.71	55.28
7	$(258, 9)$	126.80	81.51	387	142.15	126.84	62.12
6	$(322, 9)$	132.15	65.92	322	132.22	132.15	65.16
5	$(429, 10)$	150.60	50.34	429	150.60	150.60	74.63

The first-order certificate proves only that a block of dimension $r_{\min}$ can support the ranks of Ψ_1 and Ψ_2; Assumption 2 additionally assumes that this same block determines all $k_\ell - 2$ systems used for higher-order derivative recovery. At the headline $(322, 9, 6)$ cell, $r_{\min} = 375$ and full first-order rank leaves only $\dim \widehat{K}' = \binom{12}{2} = 66$ directions. The number r actually needed for higher-order derivative recovery is therefore the principal unmeasured size parameter in the Category 1 tally: it multiplies relation generation and also increases replay, map construction, recovery arithmetic, and retained state.

Table 6 reoptimizes over all admitted width-64 singleton cells under several hypothetical laws for r; these laws are sensitivities, not sufficiency claims. The original cell held fixed reaches 2^{143} near $r = 711{,}000 \approx 6.86k_\ell^2$, but the joint scan then moves to the seven-anchor cell. Consequently $6.9k_\ell^2$ moves the optimum to $(258, 9, 7)$ and gives $2^{142.19}$; the reoptimized subtotal crosses 2^{143} only at about $162k_\ell^2$ ($2^{143.00}$). Thus moderate quadratic growth preserves the arithmetic headline, but no useful target-scale upper bound or scaling law for r is known. A direct small-field test would compare the admissible sequential branches obtained from certified subblocks of increasing dimension with those from the full kernel.

Table 6. Category 1 relation-block-size sensitivity for higher-order derivative recovery. Each row reoptimizes the Affine+dense subtotal over all admitted singleton cells at width 64 under the displayed law for the required block size r. Work and active state are base-two exponents of bit operations and bits.

Assumed r	Cell	Relations	C_{rel}	Subtotal	State
$r_{\min}$	$(322, 9, 6)$	375	132.15	132.22	65.16
$2k_\ell$	$(322, 9, 6)$	644	133.02	133.06	65.93
k_ℓ^2	$(322, 9, 6)$	103,684	140.22	140.22	73.01
$6.9k_\ell^2$	$(258, 9, 7)$	459,292	137.29	142.19	72.24
$162k_\ell^2$	$(258, 9, 7)$	10,783,368	141.84	143.00	76.79

Vedenev's Conjecture 1 predicts a zero candidate-interpolation space for an incorrect support tuple rather than a gauge copy; an analogous claim is still needed for a false Frobenius-phase assignment. Let $G_{\mathrm{wr}} \le G$ be the number of semantically wrong guesses after removing the residual orbit (39). As a separate random-row benchmark, if the $R = cB$ rows of every such system were independent and uniform over $\mathbb{F}_q^U$, then a union bound over its nonzero coefficient vectors would give

$$\mathbb{E}N_{\mathrm{vec}} < G_{\mathrm{wr}} q^{-\Delta_{\mathrm{wr}}} \le G q^{-\Delta_{\mathrm{wr}}}. \tag{45}$$

Using G on the right gives conservative values. The tight displayed $c = 8$ cell gives $\log_2 \mathbb{E}N_{\mathrm{vec}} < -2482.91$, and $c = 7$ gives < -4562.49. If C_{ext} upper-bounds one complete extension, completion, and verification attempt, the same benchmark gives

$$\mathbb{E}C_{\mathrm{cont}} \le \big(N_{\mathrm{gauge}} + G_{\mathrm{wr}} q^{-\Delta_{\mathrm{wr}}}\big) C_{\mathrm{ext}}, \qquad 1 \le N_{\mathrm{gauge}} \le q - 1.$$

Here $N_{\mathrm{gauge}} = 1$ when the residual action is quotiented or the first gauge copy sent to extension succeeds; an unquotiented affine implementation may try more gauge copies if earlier fail-closed stages reject one. This is a quantitative sensitivity, not a theorem for the structured filtration rows. The bounded $m = 6$ census reported beside Assumption 5 supports the zero-wrong-leaf reading at one small parameter, while Appendix B supplies only a coarse worst-case C_{ext}.

The jointly optimal width-64 Affine+dense ladder point is $c = 6$: its $2^{132.22}$ subtotal is almost entirely the $2^{132.15}$ relation-generation term. Requesting its $r_{\min}$ relations in one width-384 batch lowers the joint optimum to $2^{129.94}$ while raising active state from $2^{65.16}$ to $2^{65.95}$ bits. The $c = 7$ row remains useful as a crossover: its $2^{142.15}$ affine subtotal is downstream-dominated, whereas finite-anchor PGL normalization and nested prefix reuse lower it to $2^{126.84}$. With the one-batch width 320, the latter becomes $2^{124.67}$ and uses $2^{63.06}$ active state bits. Those PGL+nested values depend on Assumption 3; the affine values do not use either optimization.

There is also a downstream-independent limit. Exhaustively minimizing C_{rel} over all 19,338 dimension-admitted $h = 1$ cells with $3 \le d \le 24$, modeled non-pivot weight $\lfloor k_\ell/2 \rfloor$, and direct Lucas traversal gives Table 7. Since $C_{\mathrm{acct}} \ge C_{\mathrm{rel}}$, improvements confined to normalization, phase handling, candidate interpolation, or completion cannot cross the floor for the chosen block-width model. Multi-position covers, lower-weight systematic bases, additional structure, or a different block-width model can change it.

Table 7. Exhaustive one-position relation-generation floors for mceliece348864. The tuple is (k_ℓ, d, c, b); state is active retained state.

Width model	Admitted cells	Minimizer	Relation generation	$\log_2$ state bits
Width 64	19,338	$(208, 8, 9, 64)$	112.35	54.91
One batch	19,338	$(208, 8, 9, 320)$	110.03	56.01

With phase synchronization disabled, an exhaustive search over all 19,338 admitted singleton cells with $3 \le d \le 24$, widths $b \in \{1, 2, 4, 8, 16, 32, 64\}$ plus the one-batch width of each cell, and every admissible common hold-out variant gives Table 8.

Table 8. Exhaustive mceliece348864 PGL+nested optima under active-state caps and $r_{\mathrm{BW}} = 1$. Cells are (k_ℓ, d, c). Common-kernel generation, marked †, additionally assumes common-kernel richness for higher-order derivative recovery.

$\log_2$ cap	Cell	Width	Organization	Subtotal	$\log_2$ state
53.00	–	–	–	–	no feasible cell
56.32	$(246, 8, 8)$	2	singleton	132.71	56.32
59.64	$(256, 8, 8)$	320	singleton	126.87	58.11
63.00	$(258, 9, 7)$	64	common†	124.32	62.14

The $k_\ell = 256$, $d = 8$ cell is the last admitted cell at that degree. Its surviving dimension at the costliest nested-intersection step is only 896 of $U = 229{,}376$, or 0.39%, so the capped rows place Assumption 3's generic-prefix heuristic exactly where it does the most work. The cap applies to M_b^{active}; generator-basis temporaries, access costs, and replication are outside the model. No modeled configuration fits a one-PiB (2^{53}-bit) cap. Appendix C gives the separate phase-synchronization sensitivity.

8 Potential Reductions in Relation-Generation Cost

8.1 Systematic-Basis Weight

The floor in Table 7 assigns weight $\lfloor k_\ell/2 \rfloor$ to every non-pivot column. The exact nonzero count (14) depends strongly on weight, although the row-rank charge is stable above $d + s - 1$. Table 9 holds the $c = 8$, $k_\ell = 215$, $d = 8$ cell and schedule fixed.

Table 9. Direct-traversal nonzero sensitivity to a uniform non-pivot weight at the $c = 8$ cell. Δ is the relation-generation exponent change from weight 107.

Weight	Nonzeros	Δ
60	190,035,290,622,323,575	-2.15
75	327,641,960,639,229,275	-1.36
90	525,011,004,642,895,485	-0.68
107	843,010,975,919,840,351	+0.00
120	1,171,737,009,839,400,485	+0.48
140	1,861,805,299,521,543,285	+1.14

The attacker may choose a shortening and a systematic basis, but cannot prescribe the resulting fundamental-circuit weights independently. Finding a systematic basis with lower non-pivot weights is a public matroid-basis optimization problem, plausibly attacked by ISD-style basis sampling or local basis exchanges. Before adding any savings in Table 9 to an accounted subtotal, we must measure its preprocessing cost and attainable weight distribution.

Two generic shortcuts do not yet earn a modeled factor. First, a direct restricted subset-zeta evaluation costs about wN operations per constrained point. At $w = 107$ this is 9.66 times the average Lucas-compressed sparse traversal $\nu/(mt - h)$, because the transform computes derivative levels that Lucas compression has already removed. Second, sparse structured elimination may help, as in integer-factorization matrices, but the present operator has average column degree 8,493 under the $c = 8$ uniform-weight convention. Without degree histograms and fill measurements at the target cell, importing a generic 5–20-fold dimension reduction would be unsupported.

8.2 Rank-Only Queries

Some local checks ask only for image dimensions. For any public jet block J,

$$\operatorname{rank}\begin{pmatrix} M \\ J \end{pmatrix} - \operatorname{rank} M = \operatorname{rank}(J|_{\ker M}), \tag{46}$$

and successive differences similarly give the rank of a quadratic jet restricted to the kernel of the linear jet. Black-box rank estimation can therefore replace explicit relation vectors for scalar corank tests. It does not by itself construct the image bases, zero-scheme certificate, nested derivative filtrations, or relations required by the global reconstruction. Eliminating relation batches on this basis is consequently unpriced: one must first specify the required rank or membership queries, their count, and their reliability over $\mathbb{F}_2$.

9 Asymptotic Reading

The relation-generation feasibility window compares the dimension requirement $k_\ell \gtrsim d\sqrt{mt}$ with the Hermite ceiling $k_\ell \lesssim d(2t+1)-mt$. They become compatible around $d > m/2$. At the minimal scale,

$$\log_2 N \sim \frac{m}{2}\left(\log_2 e + \frac{1}{2}\log_2(mt)\right), \qquad \log_2 C_{\mathrm{rel}} \sim m\left(\log_2 e + \frac{1}{2}\log_2(mt)\right). \tag{47}$$

When $m \asymp \log_2 n$, this is quasi-polynomial, approximately $2^{\frac{1}{2}(\log_2 n)^2}$ up to constants. It covers only relation generation. At the same scale $c \asymp 2\sqrt{mt}/d$; when also $mt = \Theta(n)$, the support and phase enumeration is $2^{\Theta(\sqrt{n})}$.

At fixed binary rate, $m \asymp \log_2 n$ and $t \asymp n/\log_2 n$. The syzygy distinguisher [14] and Briaud et al.'s conditional distinguisher bound [3, Theorem 7.2], which underlies their heuristic key-recovery method, have modeled exponent

$$\Theta\big(n(\log\log n)^3/(\log n)^2\big).$$

This grows faster than the present route's heuristic $\Theta(\sqrt{n})$, but is not a proved running-time comparison. Our route also remains non-quasi-polynomial until the support factor q^{c-f}, rather than just the phase factor m^{c-1}, is removed.

10 Conclusion

For mceliece348864, the resulting bit-operation tally is $2^{132.22}$ in the jointly optimized width-64 Affine+dense model. The explicit one-batch multiword sensitivity gives $2^{129.94}$. The corresponding one-position relation-generation floors are $2^{112.35}$ and $2^{110.03}$. On the other four parameter sets, the jointly optimized Affine+dense subtotals range from $2^{152.38}$ to $2^{154.89}$. These are conditional attack prices, not completed target-scale attacks.

The exact part of the improvement is simple: injective zero-padding gives a square operator with the same right kernel, one sparse traversal per application, and the original rank bound. Heterogeneous multiplicity and Lucas compression further reduce the relation system, while the point-block rank formula makes its dimension guarantee exact. Apon proves that the selected anchors cannot isolate a correct binary-Goppa curve and identifies full-code testing as an alternative list problem; the 26 August draft by Vedenev instantiates that route by matching unused public columns. Our continuation model follows it but remains unproved. The first experimental priority is to measure how many certified relations make the truncated systems used for higher-order derivative recovery reproduce the full-kernel branches. Further priorities are to demonstrate reliable binary block-Krylov yield, validate higher-order derivative recovery at target scale, test column matching and wrong-leaf rejection beyond the $m = 6$ experiments, and construct a compatible affine completion gauge. Lower-weight systematic bases and multi-position hold-outs are additional concrete levers.

On the Use of AI in Cryptanalysis

These results originate from a systematic effort to implement and improve cryptanalytic attacks on Post-Quantum Cryptography using AI. We used OpenAI GPT-5.6-Sol and Claude Opus 5 for theory formulation, proofs, analysis, and preparing and executing experiments. Furthermore, we used the LLMs and Grammarly to prepare and improve the presentation. The process was closely human-supervised, however, requiring hundreds of prompts and manual edits.

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A Cost-Estimate Component Status

Change note, 2026-08-27. The costing model now uses exact injective padding, active retained state, joint cell optimization, finite-anchor PGL normalization, an exact point-block-rank proof, and explicitly labelled completion and wrong-leaf heuristics.

In the following, we collect the proof, modeling, and conjectural status of every ingredient used in our cost estimation.

Used in	Ingredient and status
§3.1; Alg. 1	**Hermite soundness and heterogeneous schedule.** Proved in Lemma 1 and Proposition 1; cell arithmetic exact.
§3.2; Alg. 1	**Lucas-minimal levels.** Proved in Theorem 1 from the binary inclusion-matrix decomposition and reproduced arithmetically by the artifact model.
§3.3; Alg. 1	**Point-block rank and kernel floor.** Equation (11) is proved via the Bier–Wilson diagonal basis and independently checked on nine registered shapes and all 870 valid tuples with $k_\ell \leq 8$. Summing point ranks is an upper bound; no cross-point rank equality is claimed.
§3.3; Algs. 1–2	**Nonzero count.** Exact for the chosen public-column weights; target model uses the uniform-weight convention rather than an opened key.
§4; Alg. 2	**Padded kernel identity.** Exact by Lemma 2: when $R_M \leq N$, $B = EM$ has $\ker B = \ker M$, rank at most $\overline{\rho}$, and one sparse traversal. The augmented matrix is used only when padding is unavailable. One nominal Coppersmith attempt per batch is priced; projection failure, preconditioning, retries, and a target-scale backend remain open.
§4; Alg. 2	**Multi-position hold-out cover.** Containment, Hermite rescheduling, and direct costs exact; obtaining the same derivative filtration at every anchor from enlarged kernels is open Assumption 1.
§5; Alg. 6	**Anchor count and correct-leaf dimension.** The count $c = \max\{4, \lceil U/B \rceil\}$ models wrong-guess rejection under Vedenev's Conjecture 1 and the residual projective gauge. For the strict-profile correct binary-Goppa leaf, Eqs. (29) and (30) prove dimension at least $2t+4-c$ and at least $2t+3-c$ further correct labels [1].
§5; Alg. 6	**Per-anchor condition count.** $B = \binom{k_\ell}{2}$ is exact for an unindexed strict complete derivative filtration by Proposition 3. An indexed filtration instead contributes E_i in (32); exploiting $E_i > B$ needs a new wrong-guess rank statement.
§5; Alg. 6	**Prefix-tree nesting.** Algorithmically explicit; savings depend on the wrong-prefix dimensions in Assumption 3. The structurally larger correct prefix is not described by that assumption.

Continued on next page

Cost-Estimate Component Status (continued)

Used in	**Ingredient and status**
§5; Alg. 6	PGL_2 **normalization.** Ambient GRS covariance and preservation of every finite-anchor derivative filtration are proved in Proposition 5. Trying $c-2$ public finite values of ζ avoids the pole. Passing the resulting projective labels to an affine Goppa endpoint still requires a compatible pole/gauge.
§5; §6	**Order-basis cost.** Leading soft-O model on an unworked target-scale formulation.
§5; Algs. 3, 4	**First-order branch planes.** A bounded multiplication-matrix routine with exhaustive field-element eigenvalue scans extracts candidate lines; public quadratic forms, exact vanishing, full cross-term rank, distinct spanning quotient lines, and the Frobenius cycle certify an accepted block. The fixed uniform-form trial cap may refuse a genuine block, and constructing an accepted target-scale block remains open.
§5; Algs. 5–6	**Higher-order derivative recovery.** Algorithm 5 implements the displayed systems (25). The priced strict-profile version depends on Vedenev's Conjecture 4 and on Assumption 2's truncated-block richness; no useful bound or scaling law is known for the sufficient relation count. Table 6 reoptimizes the tally under larger hypothetical counts. General indexed profiles and plateaus are not priced.
§C; Alg. 9	**Frobenius-type decomposition.** Classical linearized-polynomial decomposition, reproved in Lemma 3 to fix conventions; synthetic tests support the balance test heuristically but are not used in the tally.
§C; Alg. 9	**Public pure cross-pairing.** Open Assumption 6. The canonical $h = 2$ jet has no mixed block. The $c-1$ two-position relation-generation computations and their replay are priced in Table 11, but the map they should feed is not constructed.
§5; App. B	**Wrong leaves and list-problem continuation.** Apon [1] identifies full-code candidate testing as a nontrivial list problem; Assumption 4 is our explicit success hypothesis for that open step. Assumption 5 is a separate generic-rank heuristic. In bounded $m = 6$ tests supplied with true secret data, all 5,748 semantic wrong profiles had full rank, while correct rank-max steps had unique correct maximizers and reached dimension one. Separately, 769 correctly labelled affine mceliece348864 points completed to an exactly verified key, but the compatible pole was supplied by calibration. Algorithms 7 and 8 are fail-closed; (52) is a conservative upper bound.
§4; §6; Alg. 2	**Memory.** Equation (22) counts the solver panel and sequence, all retained certified relations, and the sequential active peak. Generator-basis temporaries, access, communication, replication, and NIST memory penalties remain unpriced.
§5; Algs. 8, 6	**Final key verification.** Unconditional once a candidate is produced: reconstruct the complete Goppa parity-check row space and compare it with the public dual.

Accordingly, the headline $2^{132.22}$ width-64 and $2^{129.94}$ wide-block figures are heuristic end-to-end tallies, conditional on one successful binary Coppersmith attempt per relation batch and Assumptions 2, 4 and 5. Their active retained states are $2^{65.16}$ and $2^{65.95}$ bits, before generator-basis temporaries or memory penalties. Independently of downstream heuristics, the exact fixed-width relation-generation model has the $2^{112.35}$ floor in Table 7; the one-batch multi-word sensitivity lowers it to $2^{110.03}$. Phase-aligned rows add Assumption 6 and are not headline results.

B Fail-Closed Continuation

This appendix specifies the public-column continuation required by Eqs. (29) and (30). The procedure is deterministic and fail-closed for a fixed column order, but Assumption 4 is needed for it to succeed on a target-scale correct leaf. The bound below prices its work without assuming success.

Algorithm 7: ExtendLeaf() : fail-closed public-column continuation.

Input: A nonzero candidate space $W \subseteq \mathbb{F}_q[Z]^{k_\ell}_{\le D_\ell}$, assigned anchors, shortened public columns, and the original public key.
Output: An exactly verified equivalent Goppa key, or $\perp$.

foreach *unassigned shortened column* y_j *in a fixed order, while* $\dim W > 1$ **do**
  For every unused $a \in \mathbb{P}^1(\mathbb{F}_q)$, compute $W_{j,a} = \{G \in W : G(a) \in \langle y_j \rangle\}$;
  If exactly one a uniquely maximizes $\dim W_{j,a} > 0$, record it and replace W by $W_{j,a}$; otherwise skip y_j;
end
if $\dim W \neq 1$ **then**
  return $\perp$
end
Choose $0 \neq G^\star \in W$; match every shortened column to its projective value under $G^\star$ and require unique, distinct matches with nonzero multipliers and exact ambient-code containment;
return RestoreAndVerify *on the matched support and multipliers*;

For vectors $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{F}_q^s$, define

$$\mathbf{V}_r(\boldsymbol{a}, \boldsymbol{b}) = \left(b_u a_u^v\right)_{0 \le v < r,\ 1 \le u \le s}. \tag{48}$$

Let $\mathcal{I}$ contain the shortened coordinates, put $D = n - 2t - 1$, and let $\mathbf{G}_j$ generate the public code with one coordinate $j \in \mathcal{I}$ restored. Given the known support $\boldsymbol{\alpha}_{\bar{\mathcal{I}}}$ and candidate w, a multiplier witness satisfies

$$\mathbf{G}_j \operatorname{diag}(\boldsymbol{\nu}) \mathbf{V}_{n-D-1}\left((\boldsymbol{\alpha}_{\bar{\mathcal{I}}}, w), \mathbf{1}\right)^{\mathsf{T}} = \mathbf{0}. \tag{49}$$

Only a one-dimensional nullspace with an all-nonzero witness and exact code containment is accepted [15].

Algorithm 8: RestoreAndVerify() : deterministic chart restoration.

Input: A matched projective shortened support and multipliers, the public key, and the shortening map.

Output: An exactly verified equivalent Goppa key, or $\perp$.

foreach *pole outside the matched support, in a fixed order* **do**

  Transport the support and multipliers using (38); abandon the chart unless all known support values are finite;

  Restore each shortened-away coordinate by scanning every unused field element in (49); abandon the chart unless every step has one accepted value;

  Replay (49) on the full public matrix and require one all-nonzero global multiplier and exact ambient-code containment;

  Run Hemmert's support-to-Goppa completion, scanning its at most $q+1$ representatives [10];

  return *the first degree-t irreducible Goppa polynomial and distinct support whose rank is mt and whose public code is exactly the input*;

end

return $\perp$;

A tied restriction is skipped, and every column is processed only once, so Algorithm 7 terminates. For each Classic McEliece set, $n \leq q$, hence at least one projective point lies outside a complete support. A wrong chart can refuse; no refusal is interpreted as nonexistence, and only exact public-code equality authorizes output.

For a literal bound, put $K = k_\ell$, $D' = D_\ell$, $L = n_\ell$, $Q = q+1$, $U = K(D'+1)$, and $r_{\max} = \min\{K-1, U\}$. One candidate restriction costs at most

$$T_{\text{cand}} = 2D'KU + 4(K-1)Ur_{\max}. \tag{50}$$

Naive evaluation and elimination give

$$\begin{aligned} T_{\text{rm}} &\leq (L-c)\big(QT_{\text{cand}} + 2U^3\big), \quad & T_{\text{match}} &\leq 2D'KQ + 3LKQ, \\ T_{\text{restore}} &\leq 2Q\big((n-L)Q+1\big)n^4, \quad & T_{\text{comp}} &\leq 2Qn^4. \end{aligned} \tag{51}$$

Consequently,

$$\begin{aligned} C_{\text{ext}} &\leq m^2(T_{\text{rm}} + T_{\text{match}} + T_{\text{restore}} + T_{\text{comp}}) + QC^{\text{up}}_{\text{keycheck}}, \\ C_{\text{cont}} &\leq N_{\text{leaf}}C_{\text{ext}} \leq GC_{\text{ext}}. \end{aligned} \tag{52}$$

This is a finite no-success traversal envelope. It charges worst-size ranks at every candidate and says nothing useful about the structured survivor count; the main heuristic instead uses Assumption 5.

C Optional Frobenius-Phase Synchronization

The phase factor m^{c-1} in (37) is potentially removable, but no public construction presently justifies doing so. At anchor i, put $V_i = T_{0,i}/\langle y_i \rangle$. The certified pure lines determine a binary field algebra $\mathcal{E}_i \simeq \mathbb{F}_{2^m}$ without choosing its Frobenius origin. Fix a public irreducible polynomial, choose one root A_i in each representation, and define

$$\mathrm{Sol}_{ij}(\tau) = \{C \in V_i^* \otimes V_j^* : A_i^{\mathsf{T}} C = C A_j^{[\tau]}\}, \qquad \tau \in \mathbb{Z}/m\mathbb{Z}. \tag{53}$$

Lemma 3 (Frobenius-type decomposition). *Every* $\mathrm{Sol}_{ij}(\tau)$ *has binary dimension* m*, and*

$$V_i^* \otimes V_j^* = \bigoplus_{\tau \in \mathbb{Z}/m\mathbb{Z}} \mathrm{Sol}_{ij}(\tau). \tag{54}$$

Proof. Identify both modules with $E = \mathbb{F}_{2^m}$. Under the trace pairing, every binary-linear map is uniquely $L(y) = \sum_s a_s y^{2^s}$. The balance relation $L(\alpha^{2^\tau} y) = \alpha L(y)$ selects one exponent $s \equiv -\tau \pmod{m}$, with $a_s \in E$ free. The m coefficient spaces are independent and exhaust $\mathrm{End}_{\mathbb{F}_2}(E)$.

The obstruction is constructing a suitable mixed pairing. The canonical two-position order-two jet lands in $\mathrm{Sym}^2(V_i)^* \oplus \mathrm{Sym}^2(V_j)^*$, not in $V_i^* \otimes V_j^*$.

Assumption 6 (Public pure cross-pairing) *For every edge* (i, j) *of a spanning tree, a public fail-closed algorithm* CrossPair *maps* r_{pair} *certified two-position relations and the two local spaces to a nonzero* C_{ij} *lying in exactly one* $\mathrm{Sol}_{ij}(\tau)$*. Its post-generation arithmetic is lower order.*

Algorithm 9: SynchronizePhases() : conditional phase synchronization.

Construct each certified field algebra $\mathcal{E}_i$ and refuse unless it is a degree-m field;
Generate r_{pair} certified two-position relations on every edge of a spanning tree;
Call CrossPair; decompose its output by (54) and refuse unless exactly one component is nonzero;
Propagate the resulting shifts along the tree and verify every optional cycle edge;
return *relative phases modulo one global shift, or* $\perp$;

Let $(R_{\star,2}, \tau_2)$ be the rank bound and traversal count selected by (19) for the two-position cell, let ν_2 be its nonzero count, and put $n_{\mathrm{pair},b} = \lceil r_{\mathrm{pair}}/b \rceil$. Pricing one spanning tree gives

$$C_{\text{pair-rel}} = 3(c-1)\tau_2 w_b \left\lceil \frac{R_{\star,2}}{b} \right\rceil \nu_2 n_{\mathrm{pair},b} r_{\mathrm{BW}}. \tag{55}$$

Fresh annihilation and naive independence testing add

$$\begin{aligned} C_{\text{pair-replay}} &\le (c-1)\left(64 \left\lceil \frac{r_{\mathrm{pair}}}{64} \right\rceil \nu_2 + r_{\mathrm{pair}}^2 N\right), \\ C_{\mathrm{align}} &= C_{\text{pair-rel}} + C_{\text{pair-replay}}. \end{aligned} \tag{56}$$

Table 11. Frobenius-phase sensitivity for mceliece348864. Cells are (k_ℓ, d, c); work and active state are $\log_2$ bit operations and bits. “Uncharged” deletes phase guesses. “Charged” also includes the proposed two-position source, but both columns assume Assumption 6. *Order-basis rows are illustrative.

Cell	b	Solve	No sync.	Uncharged	Pair generation	Charged	State
$(215, 8, 8)$	8	Nested	135.71	118.63	118.44	119.54	54.66
$(208, 8, 9)$	16	Order basis*	140.81	116.21	–	–	54.35
$(215, 8, 8)$	32	Nested	135.71	114.79	114.52	115.66	54.94
$(208, 8, 9)$	32	Order basis*	140.81	114.51	–	–	54.57
$(215, 8, 8)$	64	Dense	144.73	119.65	112.67	119.66	55.28
$(215, 8, 8)$	64	Nested	135.71	113.13	112.67	113.92	55.28
$(208, 8, 9)$	64	Order basis*	140.81	113.25	–	–	54.91
$(258, 9, 7)$	32	Nested	128.81	128.80	128.59	129.70	61.95

At the $c = 8$, width-32 nested cell, charging the proposed relation source raises the phase-deleted subtotal from $2^{114.79}$ to $2^{115.66}$. This does not establish that saving: CrossPair is absent, and the readily available local jets have the wrong tensor type.